Atmosphere TV

DOOH Venue Incrementality & Media-Mix Measurement

A causal measurement framework, calibrated media-mix model, budget optimizer, and venue revenue + retention models — built to answer the questions Atmosphere's advertising business and its own network economics run on.

Yidan Hu

How I'd approach this role

Before the demo — the operating thesis behind it: what this role should do for Atmosphere, and how I'd prioritize the first few months as its first data scientist.

Advertising-side measurement

Prove Atmosphere's own incremental foot traffic with methods advertisers can trust — the sell-side differentiator this business runs on. (Answers Q1–Q2.)

Venue-network health

Know which venues are generating less ad revenue than their audience is worth, and which are at risk of leaving — protect and grow the inventory Atmosphere actually sells. (Answers Q3–Q4.)

Not a “quick win vs. foundation” tradeoff — do both in one move

A stratified RCT geo-holdout is the fastest trustworthy result available — and it doubles as the calibration anchor every other method here relies on: synthetic control's placebo check, the MMM's scale, and the venue models' causal-value feature. Ship it first and the “quick win” and the “foundation” are the same deliverable. The venue-network data pipeline (economics + retention features) doesn't have to wait on it — it builds in parallel.

What “done” means

Sales and account teams making different decisions with it: a trusted lift number in a client pitch, a budget conversation anchored to a real response curve, an outreach list ranked by value × risk instead of instinct — not just a model with a good accuracy number.

What follows is one concrete example of applying this approach — a synthetic demo.

Honest scope

Before the demo that follows: the scope caveats specific to this project, stated up front rather than saved for the closing slide.

Synthetic data	Every figure in this project is synthetic. The focus here is the methodology for solving the problem, not real company data or figures.
Media-effectiveness & rate-card figures	Dwell time, screen count, ad rate card, and similar parameters shown are illustrative demo values, not researched real industry benchmarks. In production: Nielsen OOH, DSP data (e.g. Vistar), or Atmosphere's own play logs and rate card.
Multi-touch attribution — deliberately not built	Atmosphere's ambient-screen model has no individual-level, cross-venue touchpoint log by default. Building MTA would require purchased mobile location/device-matching data — a real but non-default assumption, so it's scoped out rather than forced.
Online-purchase attribution — also out of scope	Every method here measures incremental foot traffic, not downstream online purchases. Tying exposure to e-commerce conversions needs a device-matched exposure-to-transaction panel — a further non-default assumption layered on top of MTA's.
Cost & revenue assumptions	The $/frequency-unit figures behind the budget allocator, and the ad-rate-card behind the venue-revenue model, are illustrative, editable placeholders — in production these come from Atmosphere's own rate card by venue type and daypart.
Retention model — its own caveats	Churn-hazard, competitive-geo, and engagement-signal parameters are illustrative, not researched attrition benchmarks. geo_cluster is deliberately excluded (small-sample tradeoff); realized_ad_revenue's importance is a venue_type confound, not a causal driver; no prospect-side retention model.

Four questions this project answers

Q1–Q2 answer the advertising side (what Atmosphere sells); Q3–Q4 answer the venue-network side (what Atmosphere runs).

1

Did the campaign actually work?

Isolate the TRUE incremental foot traffic caused by ad exposure — net of seasonality, trend, and each venue's own baseline pattern.

The sell-side differentiator: measurement Atmosphere's go-to-market team can take to market and clients can trust.

2

How should budget be spent?

Given a budget an advertiser has already committed to Atmosphere, how should it split across restaurants, gyms, bars, and waiting rooms — accounting for each venue type's own diminishing-returns curve?

An allocation Atmosphere optimizes on its own inventory and hands advertisers as part of the media plan — not a cross-channel budget call made elsewhere.

3

Where's the revenue upside?

Which existing venues are under-monetized relative to their own traffic and quality — and which prospective venues are worth prioritizing for expansion?

Atmosphere's own buy-side value question.

4

Which venues are at risk?

Which existing venues are likely to leave the network, and which ops-visible signals predict it — combined with Q3's revenue into one value-×-risk priority.

Atmosphere's own buy-side risk question.

The roadmap: four problems, one connected pipeline

Before the deep dive — this is the shape of what follows.

1 — Q1

RCT geo-holdout

High confidence

Randomized treated/holdout venues. Balance directly verified.

2 — Q1

Synthetic control

Moderate confidence

Covers historical, non-randomized campaigns. Validated with placebo tests.

3 — Q2

MMM, RCT-calibrated

Shape + scale

Adstock/saturation shape from aggregate data; scale pinned by the RCT.

4 — Q2

Budget allocator

Exact DP

Multiple-choice knapsack over calibrated response curves.

5 — Q3

Venue revenue model

GBT, honest OOF

Venue characteristics + the calibrated lift (steps 1–3) as a feature. Flags under-monetized venues; ranks expansion prospects.

6 — Q4

Venue retention model

GBT classifier, honest OOF

Ops-visible signals (engagement, uptime, complaints, outreach) + realized revenue. Flags at-risk venues; combines with Q3 into one value×risk priority.

Answers Q1 — did the campaign work?

Answers Q2 — how should budget be spent?

Answers Q3 — revenue upside?

Answers Q4 — retention risk?

One connected system, not four separate projects: the calibrated per-exposure lift from steps 1–4 becomes a feature in the venue-revenue model (step 5), which combines with the churn model's risk score (step 6) into one value-×-risk priority list.

Two data sources: a designed holdout test, and campaign history

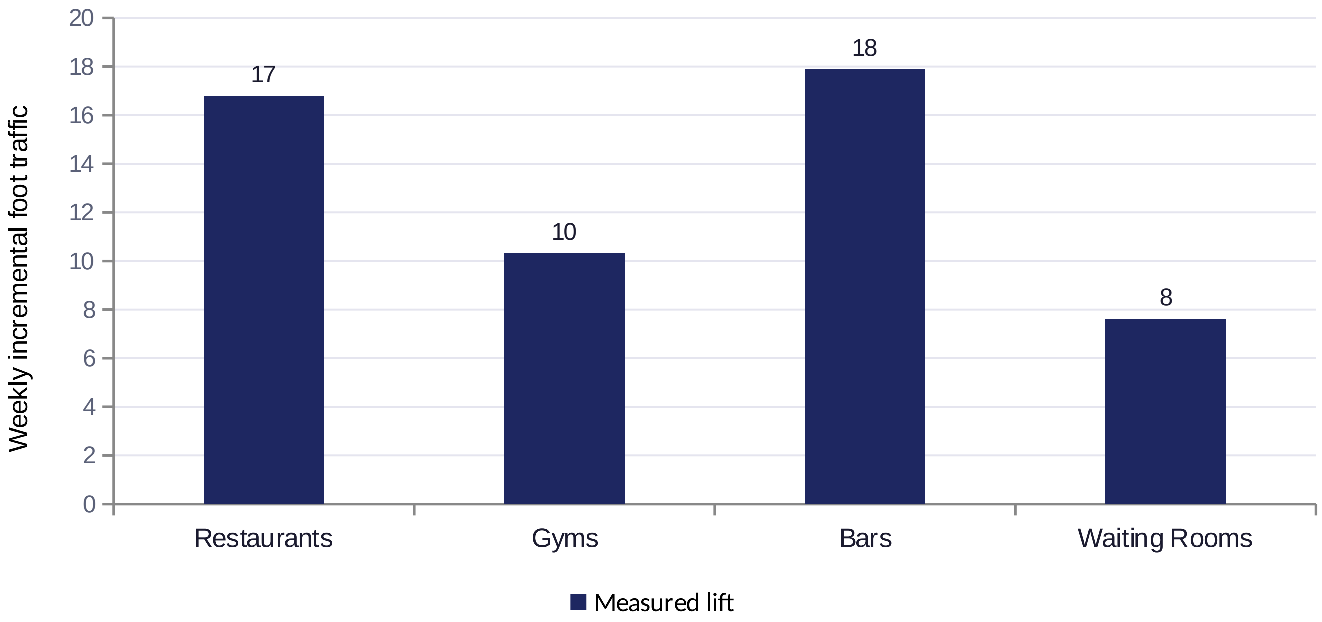
4 venue types × 100 venues each, 104 weeks (52 pre-period + 52 campaign)

Weeks	RCT pool — 136 venues	Observational pool — 264 venues
0–51 (52 wks) Pre-period	No ads for either arm — establishes each venue's own baseline.	No ads yet for anyone — identical baseline period.
52–59 (8 wks)	Still silent — the designed test hasn't launched yet. Unused by the current effect estimate.	24 of 264 go live here — each on its own timing, no coordination.
60–69 (10 wks) RCT window	Treated: fixed 6 plays/wk. Holdout: still 0. The only window the RCT estimate uses.	37 more go live here — overlap with the RCT window is coincidence, not design.
70–103 (34 wks) Post-campaign	Ads stop — a real tail holds for ~10 weeks, then fades to zero (see persistence check, next).	69 more (the largest group) go live only now — most real campaigns land late, spread thin.

RCT pool: assignment is verified, not assumed — anchors high confidence. Observational pool: 134 of 264 venues never run a campaign at all, and the rest 130 go live at their own, uncoordinated point in time — whether a venue is using its own free self-promotion slot or Atmosphere sold the remaining inventory there to an outside advertiser, the panel only records that the screen started playing ads, not who bought the airtime — exactly why synthetic control (next) reconstructs the counterfactual instead of comparing before/after directly.

The proof point: ads move real foot traffic

2/16 balance tests sig.



Venue type	n (T/H)	95% CI	p-value
Restaurants	18/17	[10.9, 22.7]	< 0.001
Gyms	15/17	[5.0, 15.7]	< 0.001
Bars	17/19	[13.1, 22.6]	< 0.001
Waiting Rooms	17/16	[5.0, 10.3]	< 0.001

Why 2/16 isn't a red flag

16 t-tests = 4 pre-treatment covariates (baseline traffic level, dwell time, screen count, audience quality) × 4 venue types — a separate check from the 136 RCT venues themselves. Only 2 came back significant at p≤0.05.

If randomization is clean, the count of false positives across 16 independent t-tests follows Binomial(16, 0.05): mean 0.8, P(≥2) ≈ 19%. Two sits well inside that distribution's main mass — nowhere near a red-flag tail value like 8+. Randomization worked as designed, not assumed.

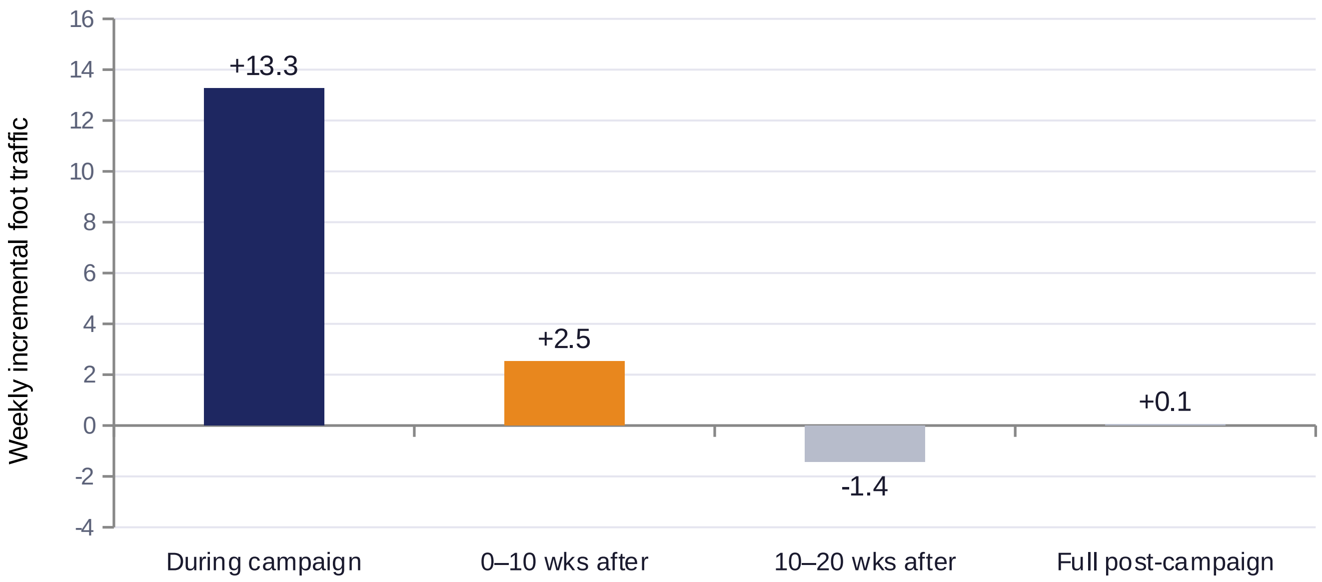
Method: ANCOVA-style delta estimator

① Per-venue delta = campaign-window avg − pre-period avg — nets out that venue's own baseline.
② Effect = mean(treated deltas) − mean(holdout deltas) — a DiD-style comparison; any trend common to both arms cancels out too.
③ Welch's t-test on the two delta samples (unequal variance, unequal n) → SE, 95% CI, p-value.

Why delta, not raw post-period levels: individual venues differ a lot in baseline traffic even within one arm — differencing each against its own pre-period average removes that noise, so 15–19 venues/arm reach significance a raw-level comparison couldn't. Why Welch's, not Student's or z: arms are small and unequal in size, so we don't assume equal variance, and we use the t- (not z-) distribution — for both the p-value and the CI — because that variance is itself estimated from so few venues.

How long does the lift last after the campaign ends?

Same treated-vs-holdout contrast as the previous slide, applied to the weeks after the campaign ends — no new design, no new assumption.



n = 67 treated / 69 holdout (same venues, every window)

Window	95% CI	p-value
During campaign	[10.8, 15.8]	< 0.001
0–10 wks after	[0.2, 4.8]	0.031
10–20 wks after	[-3.8, 0.9]	0.236
Full post (34wk)	[-1.4, 1.5]	0.945

A small tail, then nothing

The first 10 weeks after the campaign ends still show a real, significant tail — about 19% of the in-campaign lift (p=0.031). By 10–20 weeks out it's gone (p=0.236), and averaged across the full 34-week post-campaign panel the effect is a clean +0.05 (p=0.945).

Practical read: don't plan on meaningful carryover past the campaign's own window — each flight buys its own 10 weeks of impact, not lingering awareness. That argues for cadence, not one-and-done bursts.

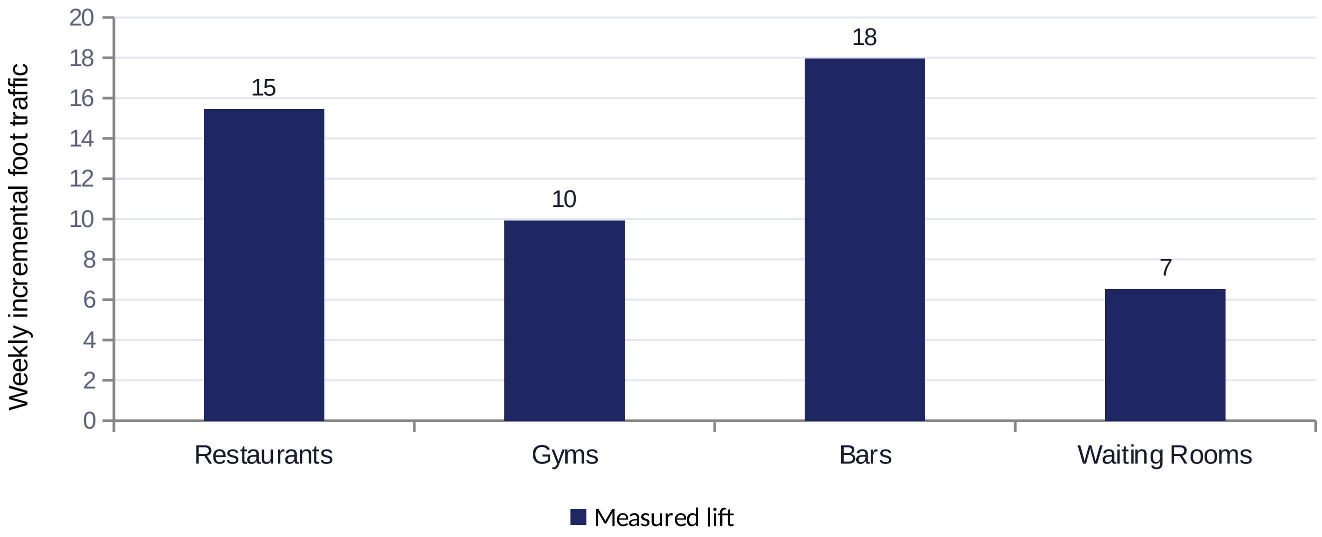
Method: same estimator, later windows

Same 3-step ANCOVA-style delta as the RCT slide — per-venue delta (window avg − pre-period avg) → treated-mean minus holdout-mean → Welch's t-test for SE, CI, p-value — but pooled across all 4 venue types here (not split like the RCT slide), and re-applied to each window instead of the single week 60–69 window.

Splitting was already thin at ~15–19 venues/arm for one window; spreading that across four venue types and four windows, on top of decay effects that are smaller than the in-campaign lift, would push the noise past what a week-level estimate can resolve.

Measuring the campaigns we couldn't randomize

Most advertisers' campaigns already ran, on venues they picked — not us. Synthetic control reconstructs the counterfactual anyway, with its own built-in checks.



Method: build a synthetic twin, per venue

① Donor pool = same-type venues that never ran a campaign (e.g. 31 never-activated bars).
② Fit non-negative weights, one per donor, summing to 1, via SLSQP (Sequential Least Squares Programming, scipy.optimize) minimizing the MSE between the weighted donor blend and this venue's own pre-period traffic (Abadie et al.) — a real blend, typically 5–15 donors get meaningful weight, not one single match.
③ Apply that same weight vector to the donors' post-period traffic → synthetic counterfactual; effect = mean(actual − synthetic) over the venue's own 16-week post-campaign window.

Unlike the RCT, each venue picks its own pre/post split at its own activation week — no shared campaign window. Each bar is the mean effect across only that type's good pre-period-fit venues (see right) — poor fits are excluded, not averaged in.

Venue type	n(used/flag)	Placebo p	RMSPE
Restaurants	42/4	0.158	26.4
Gyms	31/0	0.054	14.4
Bars	29/4	0.097	17.8
Waiting Rooms	20/0	0.106	8.3

Pre-period RMSPE (root mean squared prediction error) = √mean((actual − synthetic)²) over each venue's own pre-period weeks — how well the twin matches reality before treatment starts. Flagged (excluded from the bar) if RMSPE exceeds 2.5× its type's median — restaurants' high median (26.4) is why 4 of its 46 fits got flagged.

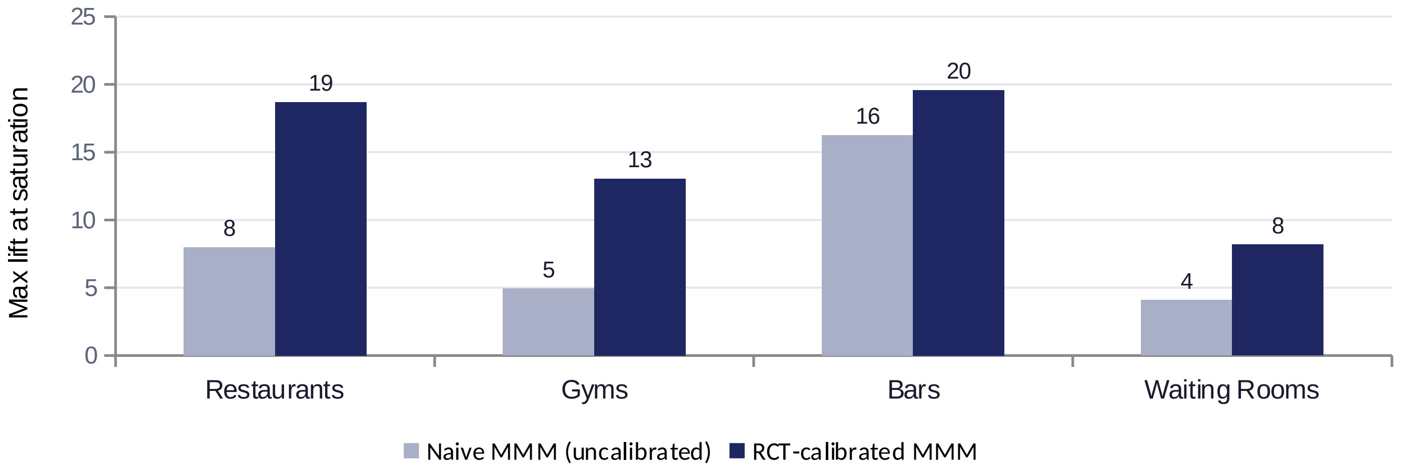
In-space placebo test — substitutes for a p-value since synthetic control has no closed-form SE: rerun the identical fit on every never-activated donor, pretending each was treated. Placebo p = share of donor "fake effects" at least as extreme as the real one — restaurant's p=0.158 is the weakest signal (its worst-fitting type too); gym's p=0.054 the strongest.

Consistent with the RCT — not independently significant

No type clears $p<0.05$ alone (gyms closest, 0.054) — but every point estimate lands within ~1.3 of the RCT's own effect for that same type, on a completely different set of venues: real corroboration, just not from this test alone.

MMM: an uncalibrated model would have underpriced results

Adstock + saturation shape comes from the aggregate weekly series; scale is pinned by the RCT's high-confidence estimate.



Calibration adjustment

Restaurants	+134%
Gyms	+165%
Bars	+20%
Waiting Rooms	+100%

Method: fit the shape, then pin the scale

① Fit shape on the aggregate weekly series (all venues of this type, RCT + observational pooled together): adstock carries exposure forward week to week (adstock_t = freq_t + decay×adstock_{t−1}), then a Hill curve turns that into diminishing returns (sat(x) = x^shape / (x^shape + half^shape)). decay, half, and shape are grid-searched — 7 × 6 × 4 = 168 combinations — picking whichever minimizes squared error against actual weekly traffic (with trend + seasonality controls, plain OLS). This gives β_naive, the raw uncalibrated max-lift.
② Pin scale only, via the RCT: simulate the RCT's own exposure (6 plays/wk for 10 weeks) through this same fitted decay/saturation curve to get its implied saturation level, then rescale β so the model's predicted lift there matches the RCT's point estimate exactly. decay/half/shape stay exactly as fit in step ① — only β moves.

Simplified for the demo: a real deployment would add more controls (competitor spend, price/promo, macro trend), validate the shape out-of-time rather than on in-sample SSE alone, and often blend the experiment in as a Bayesian prior rather than a hard rescale.

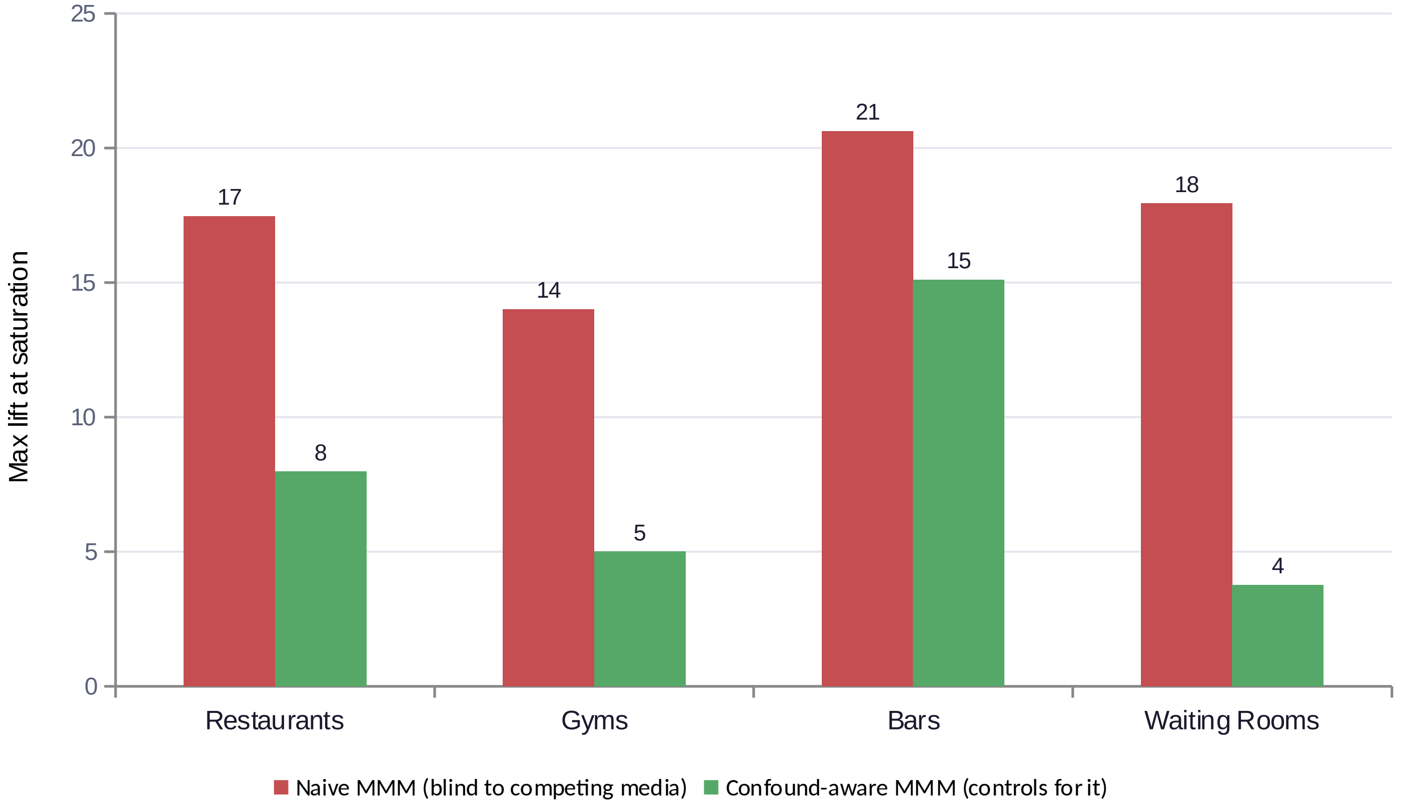
Recovery vs. true effect (synthetic ground truth)

Venue type	Naive err	Calib err
Restaurants	-10.0	+0.7
Gyms	-9.1	-0.9
Bars	-5.8	-2.4
Waiting Rooms	-4.9	-0.8

Every calibrated error is far smaller than its naive error (restaurant: −10.0 → +0.7) — only checkable here because this is synthetic data with a known answer key. In a real deployment there's no such key; that's exactly why the RCT anchor is trusted rather than verified.

Robustness check: does competing media break this MMM?

A stress-test scenario: a competing advertiser ramps up on the same venues at the same time, correlated with Atmosphere's own exposure — the kind of overlap a real MMM has to prove it's robust to.



RCT estimate shift under the identical shock

0.00

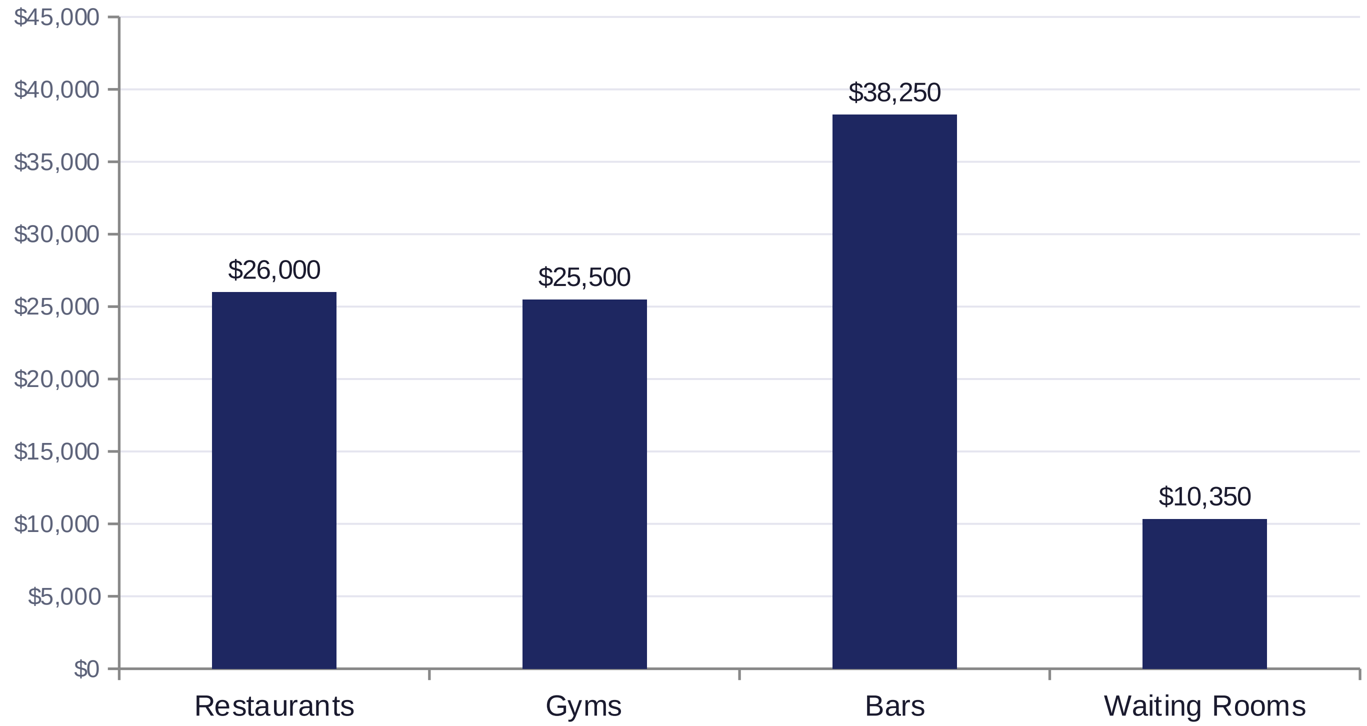
every venue type — the shock hits both arms equally, so a randomized comparison is mechanically immune to it

Naive MMM, by contrast, swings from useful to actively misleading: for 3 of 4 venue types the confound-blind fit was off by 2–4× — even at a moderate 0.35 correlation between competing media and Atmosphere's own exposure ramp, nowhere near a worst case. Controlling for the confound (green) recovers this model's own baseline from the previous slide — the RCT-scale calibration from that same slide is what closes the rest of the gap.

Why this is scoped this way: deciding how an advertiser splits budget across Atmosphere + TV + social is the advertiser's/agency's call — Atmosphere has no visibility into competitors' channel data anyway. What Atmosphere's DS role should own is netting competing media OUT of its own causal read (the JD's own phrasing), which the RCT does by design and a single-channel MMM only does with an explicit control.

Budget allocator: exact DP, not a greedy walk

A Hill/S-shaped response curve is CONVEX before its inflection point — a greedy "spend the next dollar on whichever venue type currently looks best" heuristic isn't guaranteed optimal there. An earlier greedy version of this allocator actually underperformed a naive equal-split baseline; the DP formulation below has no concavity requirement and is guaranteed to find the grid-optimal allocation.



$100,000 weekly budget example

$20,000 budget **+48.6%**

vs. naive equal-split

$50,000 budget **+11.1%**

vs. naive equal-split

$100,000 budget **+2.4%**

vs. naive equal-split

The optimizer's edge is largest when budget is scarce — exactly when allocation decisions matter most.

The other side of the business

One system answers the advertising side. The same system answers the venue-network side too.

Advertising incrementality

- Prove and price advertising incrementality
- Feeds go-to-market: sell-side differentiator, client trust
- RCT + synthetic control + calibrated MMM + budget DP

Feeds the venue-economics model as a validated causal-value input (calibrated per-exposure lift), not a separate silo.

Venue network economics

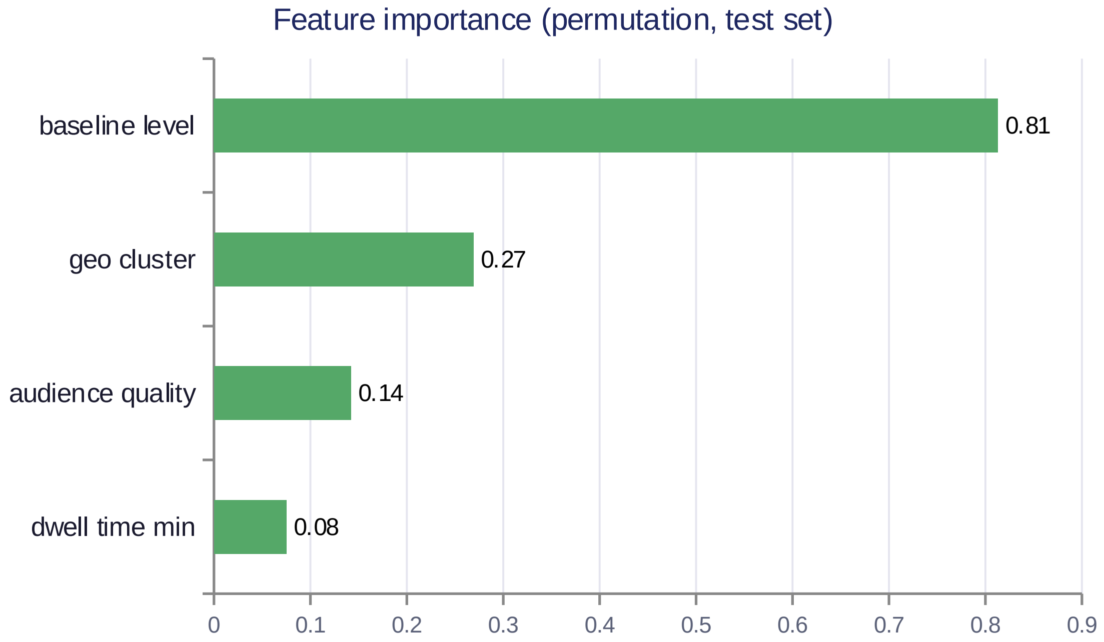
- Predict realized ad-revenue (Q3) and 90-day churn risk (Q4) from characteristics and ops signals
- Flag under-monetized (Q3) and at-risk (Q4) existing venues for sales/ops follow-up
- Rank prospective venues for expansion priority (Q3)
- Combine Q3 revenue + Q4 risk into one value-×-risk priority quadrant
- GBT regressor + classifier, honest out-of-fold evaluation throughout

“Smarter on both sides of the business” — the venue-side model shares its data foundation and causal-value input with the advertising side, rather than being a disconnected second project. Results on the next slide.

Predicting venue ad-revenue, honestly

Gradient-boosted trees on observable venue characteristics + the calibrated per-exposure lift as a feature.

0.84	22.1%	$45	20
Held-out R^2	Held-out MAPE	Held-out MAE	Venues flagged



Top under-monetized venues (of 20 flagged)

Venue #85 (Restaurants)	-62%
Venue #309 (Waiting Rooms)	-61%
Venue #53 (Restaurants)	-61%
Venue #247 (Bars)	-60%

Top prospect venues (expansion priority)

P0018 — Bars (existing market)	$684
P0055 — Restaurants (existing market)	$524
P0000 — Bars (new market)	$500
P0001 — Bars (new market)	$496

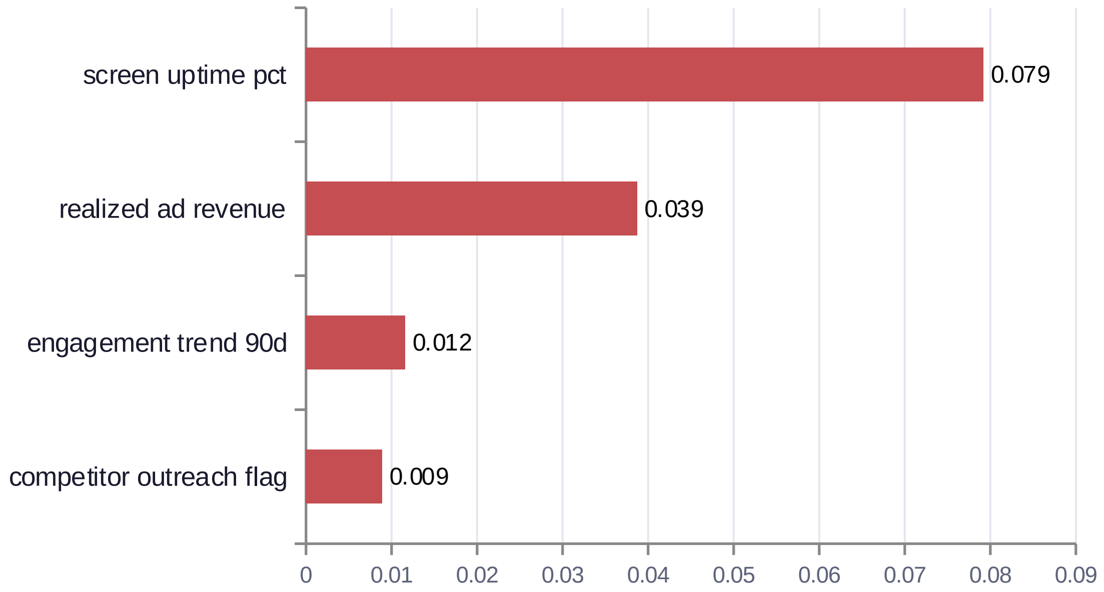
Flags come from 5-fold out-of-fold predictions, never a model scoring the venue it was trained on. The calibrated-lift feature carries near-zero importance — honest, since it's constant within venue_type once venue_type itself is a feature.

Predicting venue churn risk, honestly

Gradient-boosted classifier on ops-visible signals (engagement, uptime, complaints, competitor outreach) — the other half of "what makes a venue valuable and what puts it at risk."

0.65	**0.27**	**0.13**	**20**
Held-out AUC	Held-out PR-AUC	Brier score	Venues flagged

Feature importance (permutation, test set) — leading indicators



Top at-risk venues (of 20 flagged, by OOF risk)

Venue #281 (Bars) — 60%
engagement declining → content/placement review

Venue #200 (Bars) — 55%
engagement declining → content/placement review

Venue #12 (Restaurants) — 52%
uptime issue → dispatch technical support

Priority quadrant (Q3 revenue × Q4 risk)

Save now (high value, high risk)	136
Monitor (low value, low risk)	136
Protect (high value, low risk)	64
Low priority (low value, high risk)	64

Flags come from 5-fold OOF predictions; AUC/PR-AUC read modest next to the revenue model's R^2 because churn is a rarer, noisier event to call than revenue — typical for a churn classifier, not a modeling miss. realized_ad_revenue's importance is a venue_type confound, not a causal driver. Each "Save now" / flagged venue also carries a SHAP top driver (from the fold model that never saw it) mapped to a concrete action — technical dispatch, AM retention call, content review — not a generic "reach out."

Takeaways

1. Match the method to the assignment mechanism — RCT where randomization is designed, synthetic control where it isn't, each with its own validation check.

2. An uncalibrated MMM understated the RCT-calibrated estimate by up to 165% here — experiment calibration isn't optional polish, it's the identification fix.

3. Optimization needs the right algorithm for the curve's shape — a greedy heuristic silently lost to a naive baseline on non-concave response curves; the exact DP formulation doesn't.

4. One connected system, not four separate projects: the venue-revenue model reuses the causal pipeline's per-exposure value as an input; the retention model then combines its own churn-risk score with that revenue prediction into one value-×-risk priority quadrant — both models are evaluated honestly out-of-fold, never scored on a venue they were trained on, before either is trusted.

Thank you